PARTITIONED REGRESSION

Fundamentals of

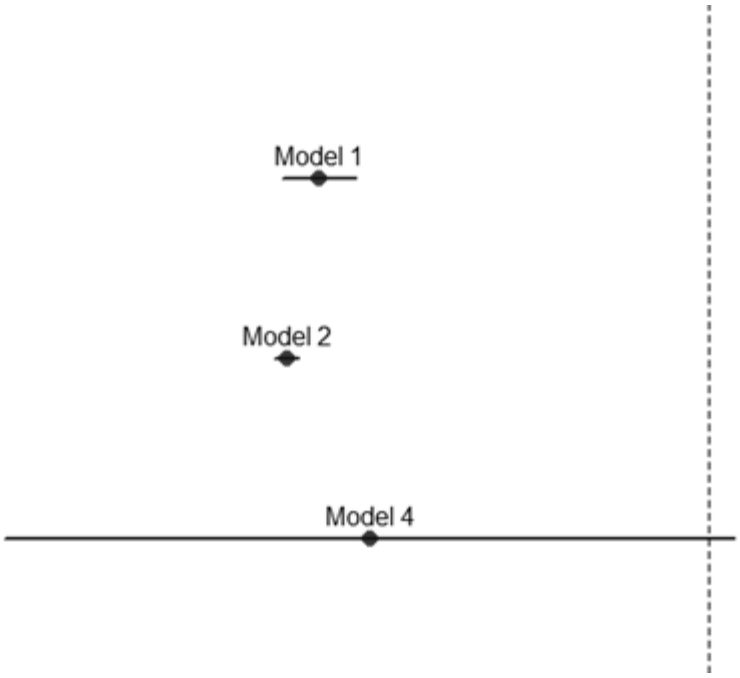
PROGRAM EVALUATION

JESSE LECY

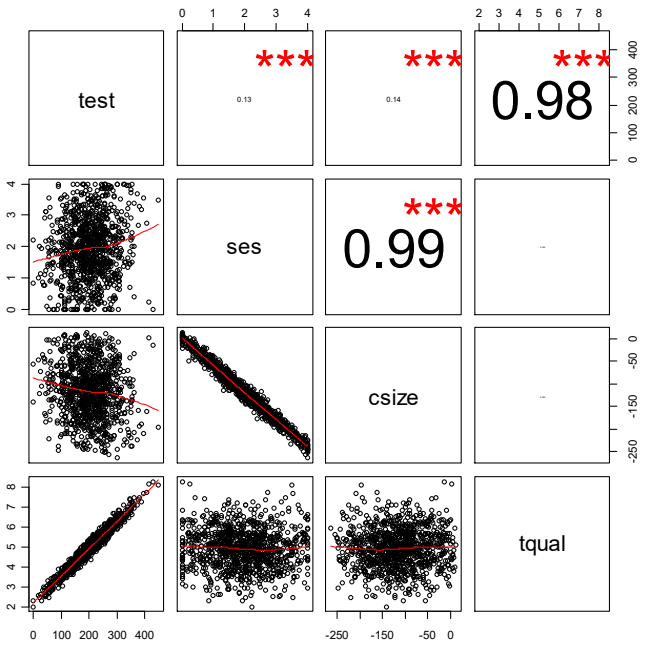
EXAMPLE: CLASSROOM SIZE

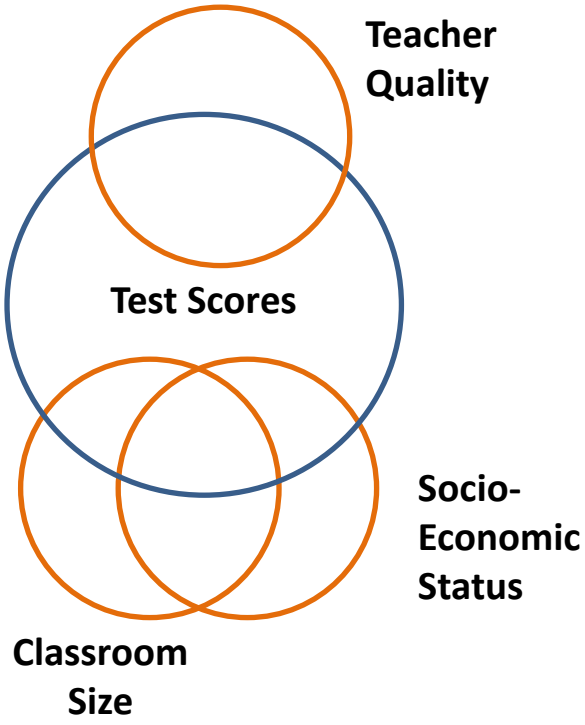
	Model 1	Model 2	Model 3	Model 4	Model 5
Constant	141.675***	−179.058***	141.479***	141.553***	−179.725***
	(4.571)	(1.609)	(4.601)	(4.596)	(1.575)
Class Size	−0.433***	−0.468***		−0.377	−0.267***
	(0.021)	(0.003)		(0.207)	(0.029)
Quality of Instruction		62.125***			62.169***
		(0.285)			(0.278)
Socio-Economic Status			43.420***	5.649	20.344***
			(2.076)	(20.853)	(2.918)
R-squared	0.307	0.986	0.305	0.307	0.986
N	1000	1000	1000	1000	1000

Why are slopes and standard errors changing when we add “control” variables?

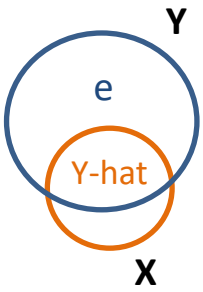


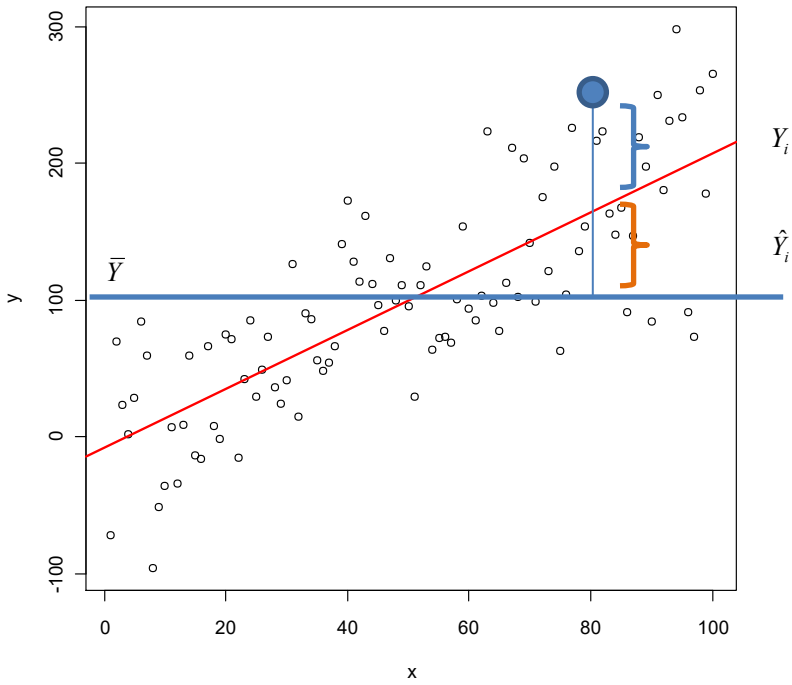
EXAMPLE: CLASSROOM SIZE



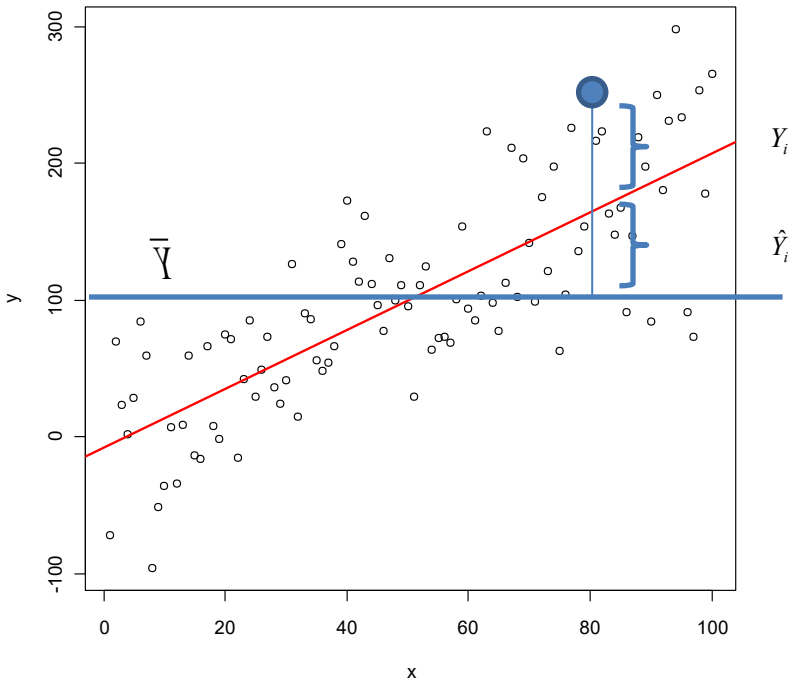


PARTITIONING THE VARIANCE OF Y





STANDARD ERROR IN REGRESSION



$$SSE = \sum e_i^2$$

Sum of Squared Error Terms

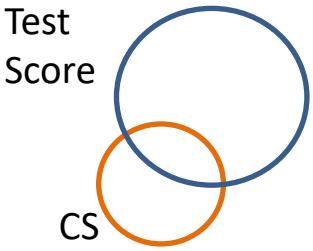
$$\hat{\sigma}_\varepsilon^2 = \frac{SSE}{n-2}$$

Variance of the residual

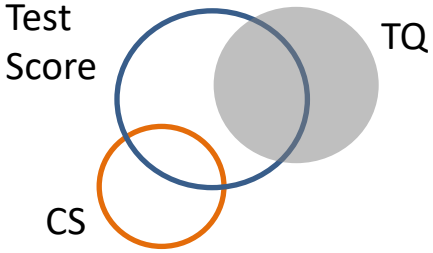
$$SE_{b_1} = \sqrt{\frac{\hat{\sigma}_\varepsilon^2}{\sum (x_i - \bar{x})^2}}$$

Standard error of the slope

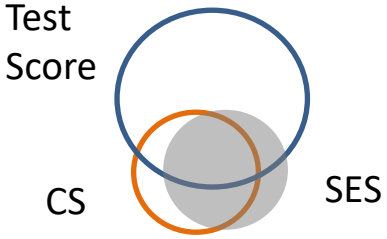
Model 1

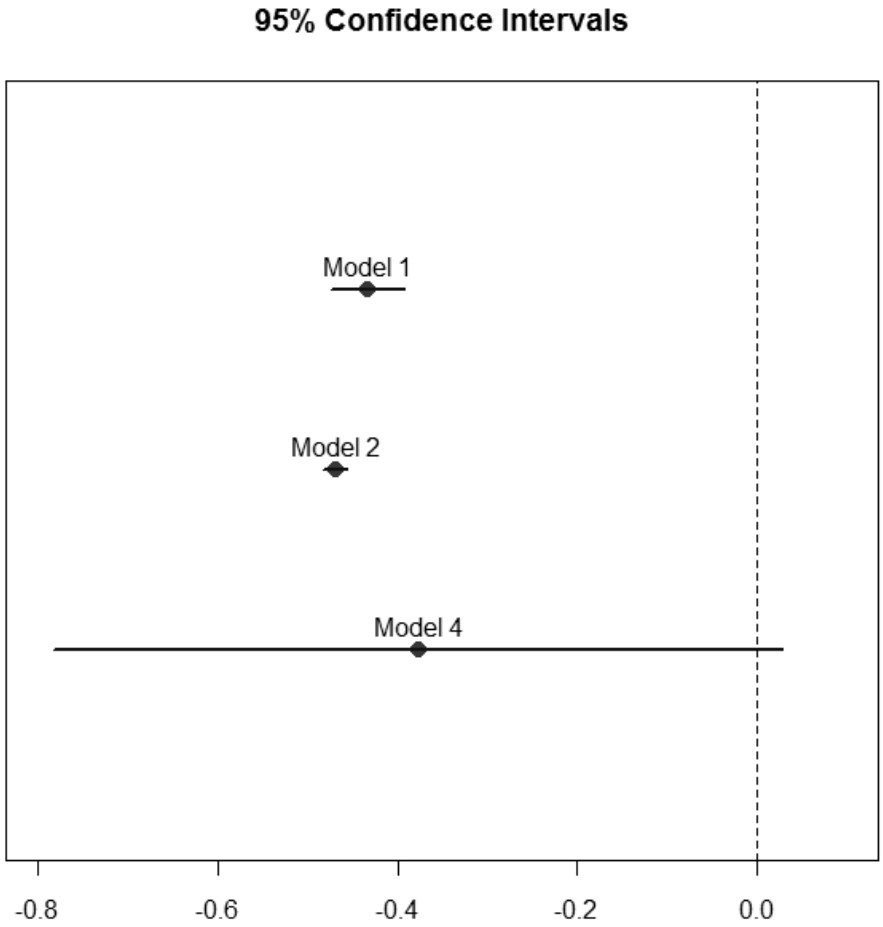


Model 2



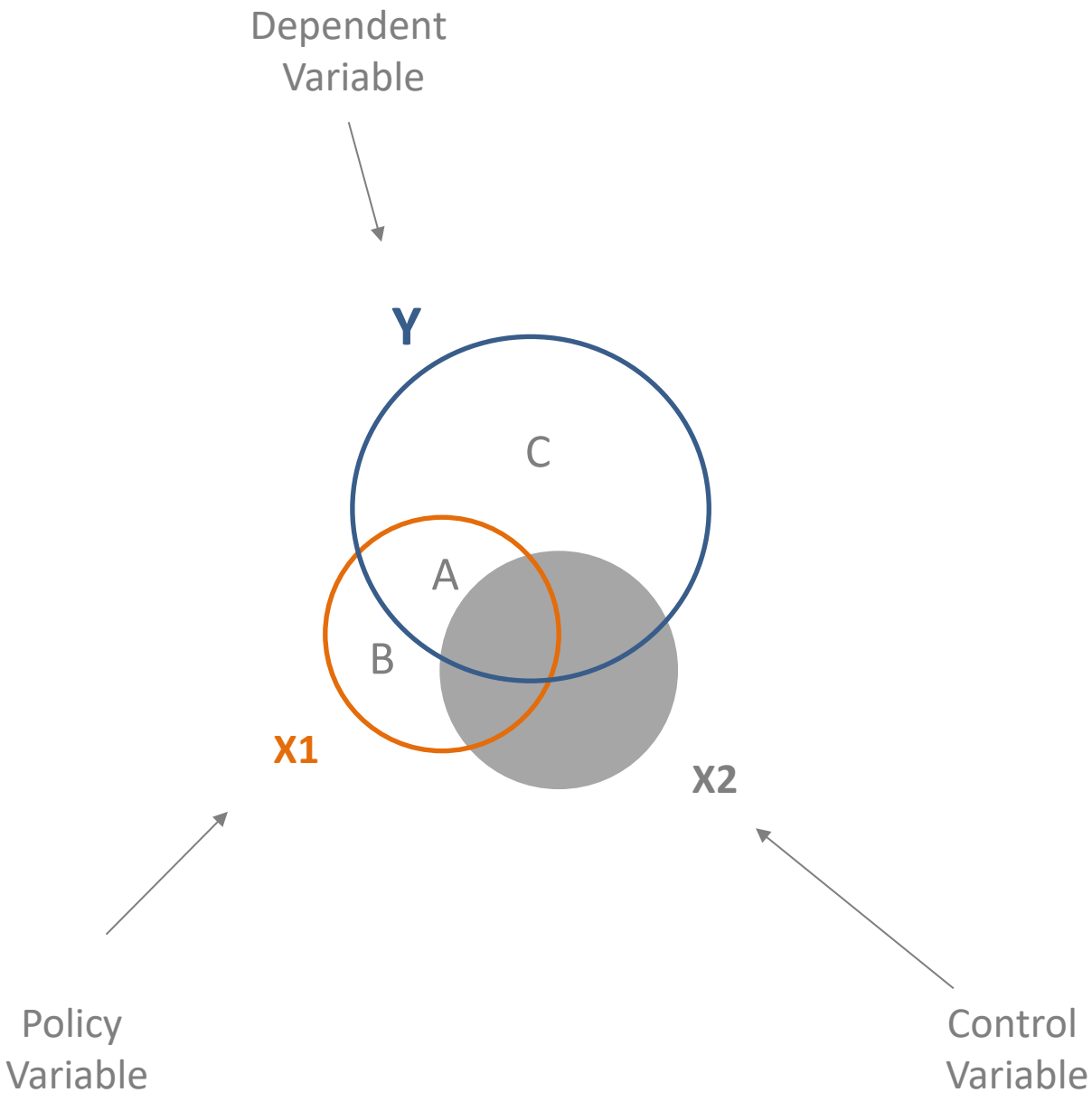
Model 4



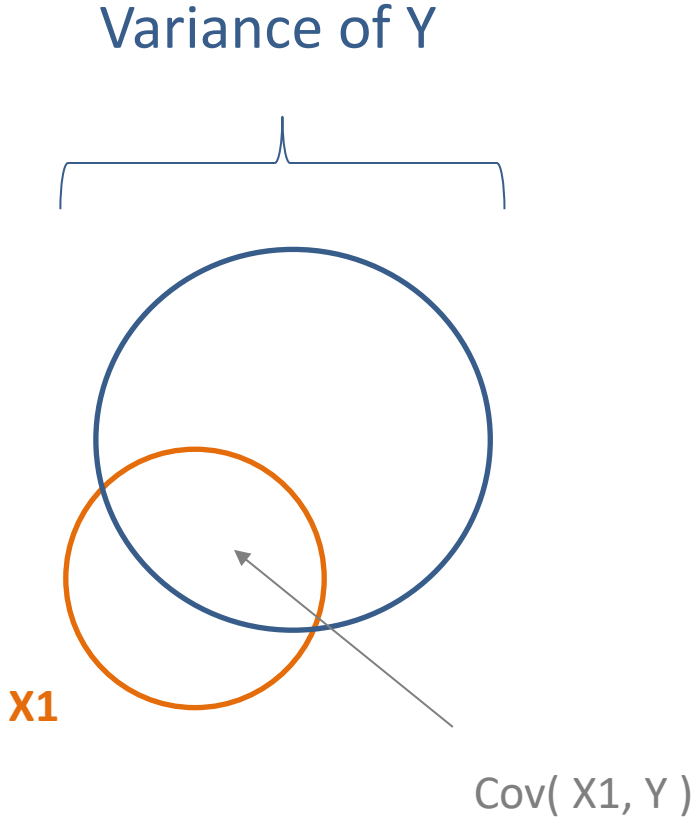


PARTITIONED REGRESSION

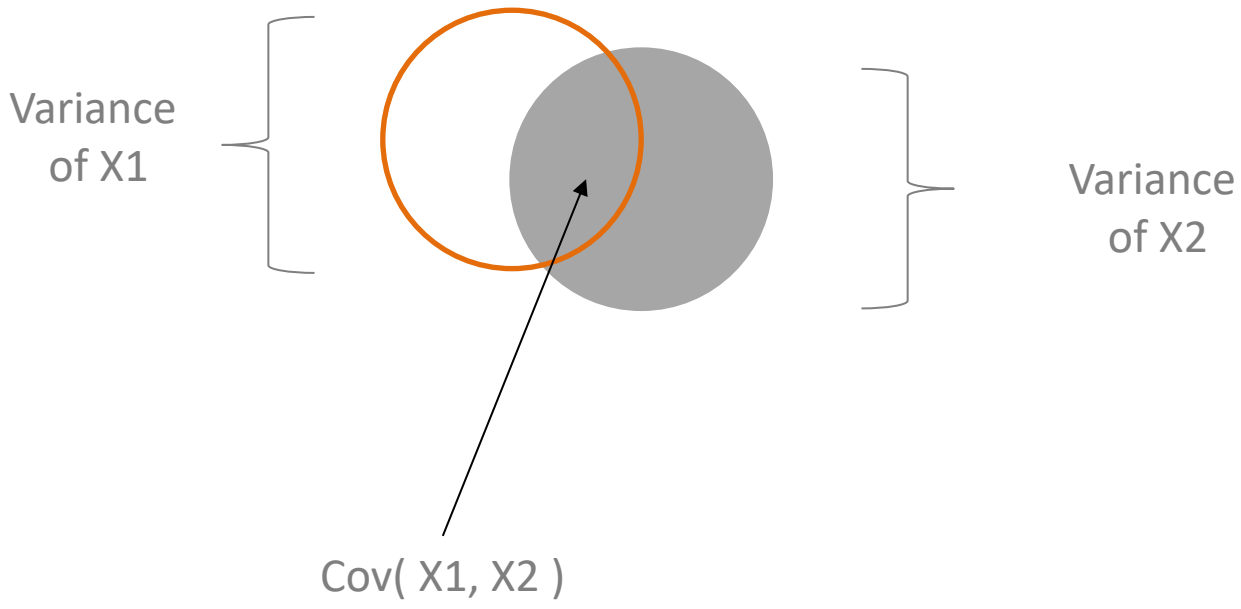
BALLENTINE VENN DIAGRAM



BALLENTINE VENN DIAGRAM



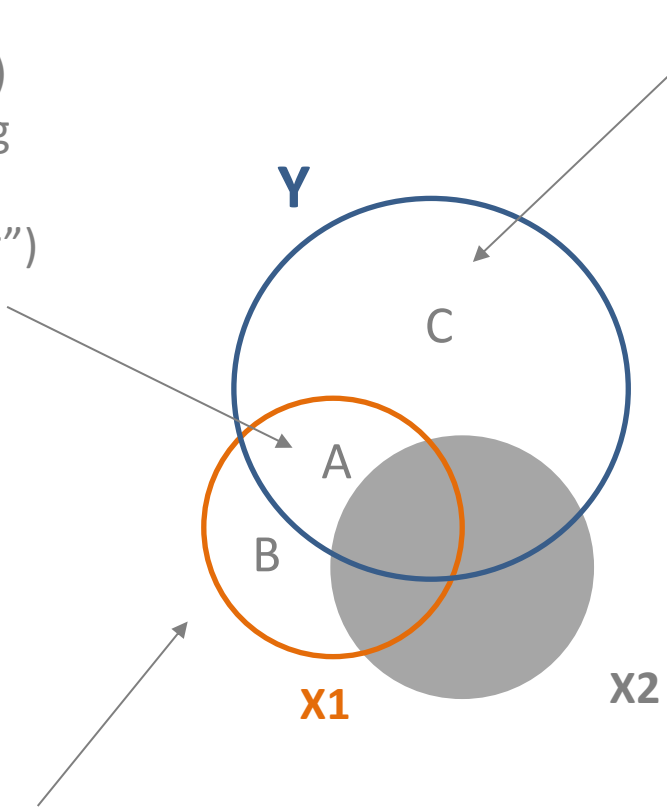
BALLENTINE VENN DIAGRAM



BALLENTINE VENN DIAGRAM

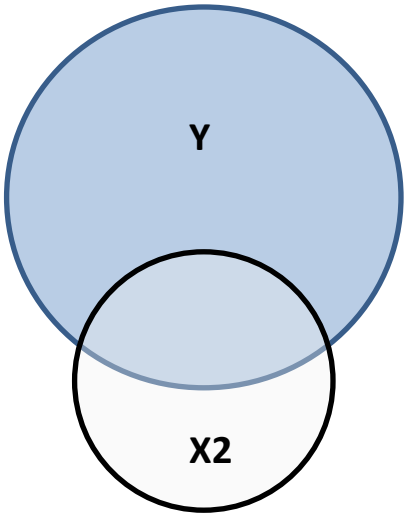
C = Unexplained Portion of Y
(the residual)

A = Cov(X1, Y)
after removing
effects of X2
(“controlling for”)

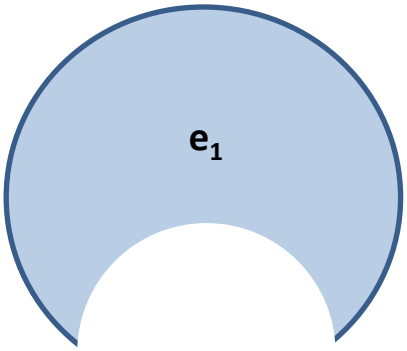


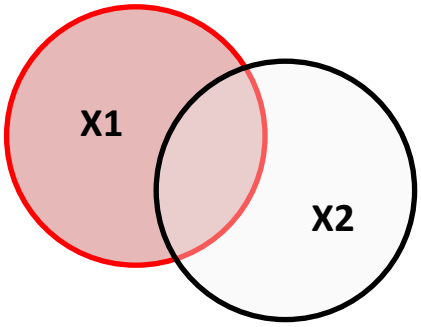
A + B = Variance of X1
after controlling for X2

PARTITIONED REGRESSION

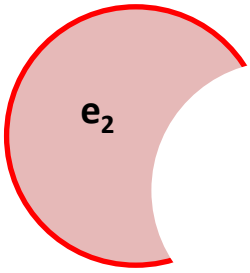


$$Y = a_0 + a_1 X_2 + e_1$$

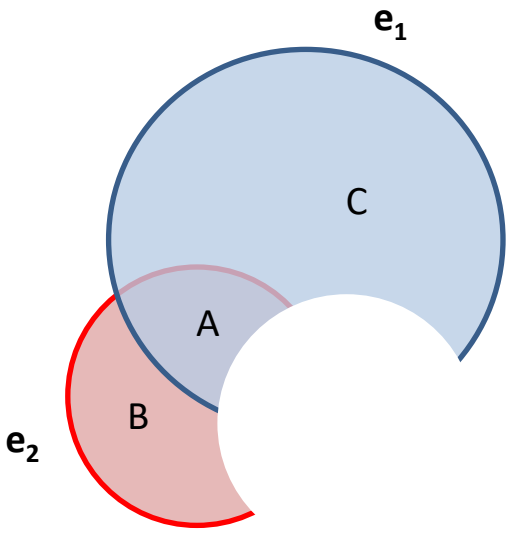




$$X_1 = b_0 + b_1 X_2 + e_2$$



PARTITIONED REGRESSION



$$e_1 = \alpha_0 + \beta_1 e_2 + \varepsilon$$

$$\beta_1 = \frac{A}{A+B}$$

$$\varepsilon = C$$

WHY CAN WE DO THIS?

X	Y	$\hat{Y} = 0.6 + 0.9x$	e	e^2
2	3	0.6+0.9*2 = 2.4	.6	.36
4	3	4.2	-1.2	1.44
6	6	6	0	0
8	9	7.8	1.2	1.44
10	9	9.6	-.6	.36
Mean=6	Mean=6	Mean=6	Sum=0	Total= **3.6**

The residual is just another variable.

The order of operations is important in partitioned regression.

NOTE: MULTIPLE REGRESSION CONTEXT

$$Y = a_0 + a_1 X_2 \ldots + a_k X_k + e_1$$

$$X_1 = b_0 + b_1 X_2 \ldots + b_k X_k + e_2$$

If there is more than one control variable make sure you cleanse all of the nuisance variance from both your dependent variable and your policy variable.

EXAMPLE 1:

$$SWB = \beta_0 + \beta_1 \boldsymbol{PSS} + \beta_2 ND + \beta_3 NP + \varepsilon$$

"policy variable"

Coefficients[a]						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-1.507	2.633		-.572	.567
	Perceived Social Support	**.310**	.032	.460	9.645	.000
	Network Diversity	-.215	.210	-.055	-1.025	.306
	# People in Social Network	.023	.014	.087	1.654	.099

Dependent Variable: **Subjective Well Being**

AUXILIARY REGRESSION #1

$$SWB = b_0 + b_1 ND + b_2 NP + e_1$$

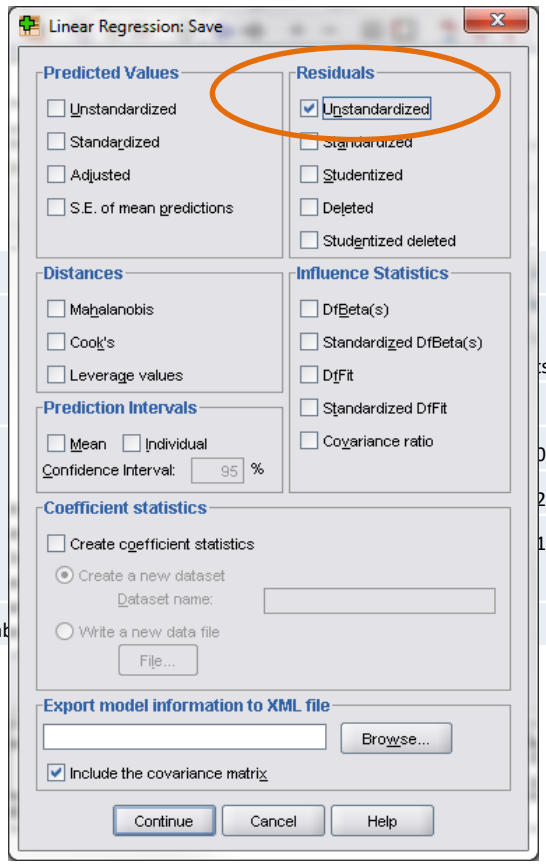
Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	21.630	1.207		17.915	.000
	Network Diversity	.224	.228	.057	.979	.328
	# People in Social Network	.039	.016	.148	2.540	.011

Dependent Variable: **Subjective Well Being**

Regress the DV on all IV's except the policy variable.
Save the residuals.

AUXILIARY REGRESSION #1



$+ e_1$

Model		Standardized Coefficients Beta	t	Sig.
1			17.915	.000
		.057	.979	.328
		.148	2.540	.011

a. Dependent Variab

AUXILIARY REGRESSION #2

$$PSS = a_0 + a_1 ND + a_2 NP + e_2$$

Coefficients[a]						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	74.720	1.720		43.438	.000
	Network Diversity	1.417	.325	.244	4.356	.000
	# People in Social Network	.052	.022	.133	2.364	.019

Dependent Variable: **Perceived Social Support**

Regress the policy variable on all of the IV's.
Save the residuals.

PARTITIONED REGRESSION

$$e_1 = c_0 + \beta_1 e_2 + \varepsilon$$

Regress the residuals on each other.

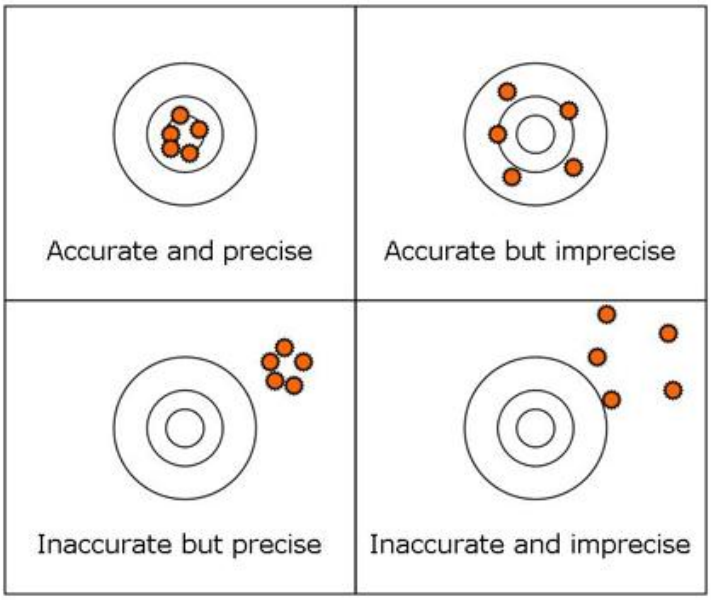
Coefficients[a]						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	2.369E-15	.253		.000	1.000
	Unstandardized Residual	**.310**	.032	.441	9.670	.000
Dependent Variable: **Unstandardized Residual**						

We have recovered the original slope!

WHY IS THIS IMPORTANT?

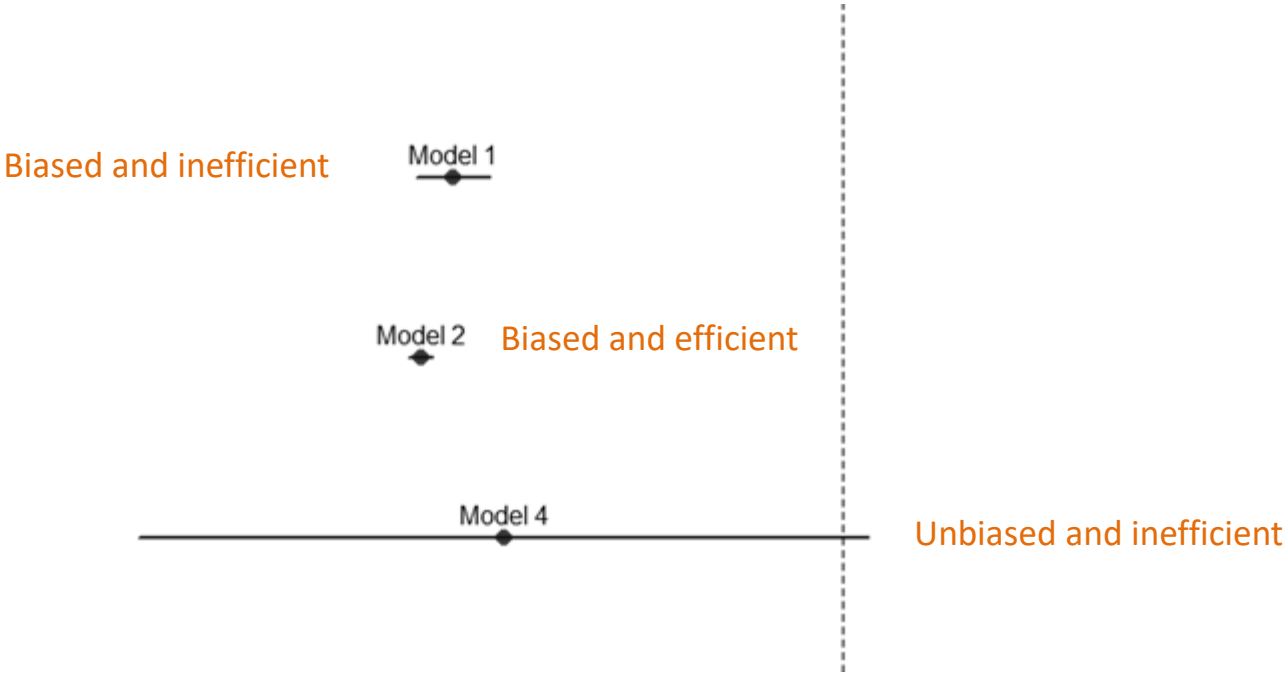
- It illustrates the idea of statistical controls in policy work – get rid of the nuisance variance!
- Provides insight into the regression error term.
- Serves as a tool for future methods (instrumental variables).
- It paved the way for modern computational econometrics.

WHY WE CARE:

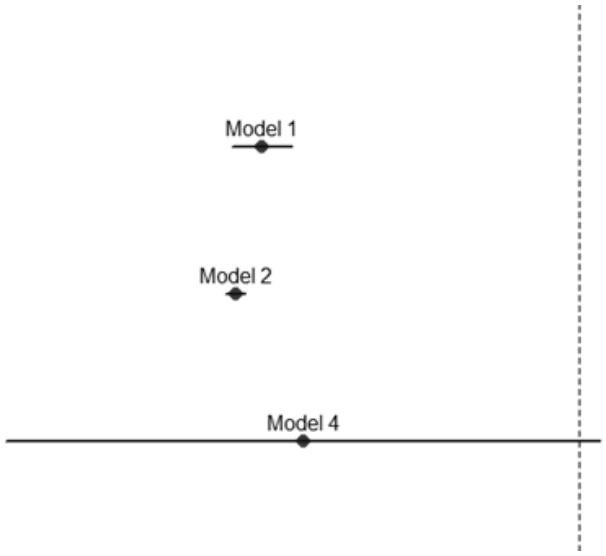


Results need to be:

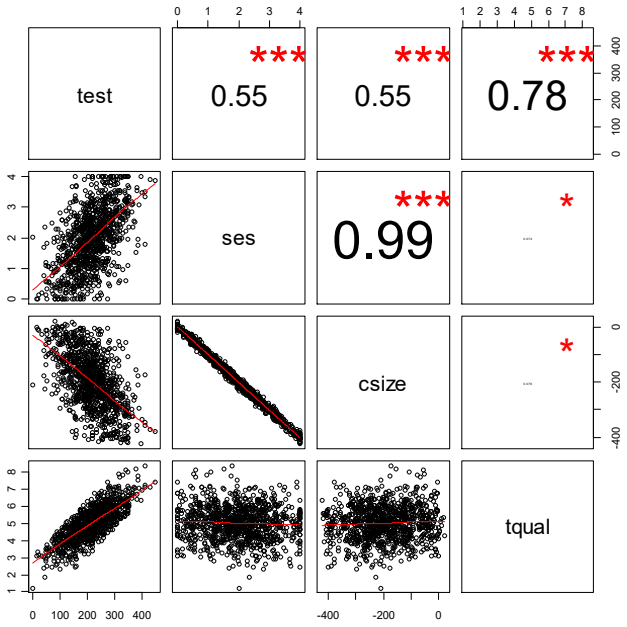
- Unbiased (accurate)
- Efficient (precise)



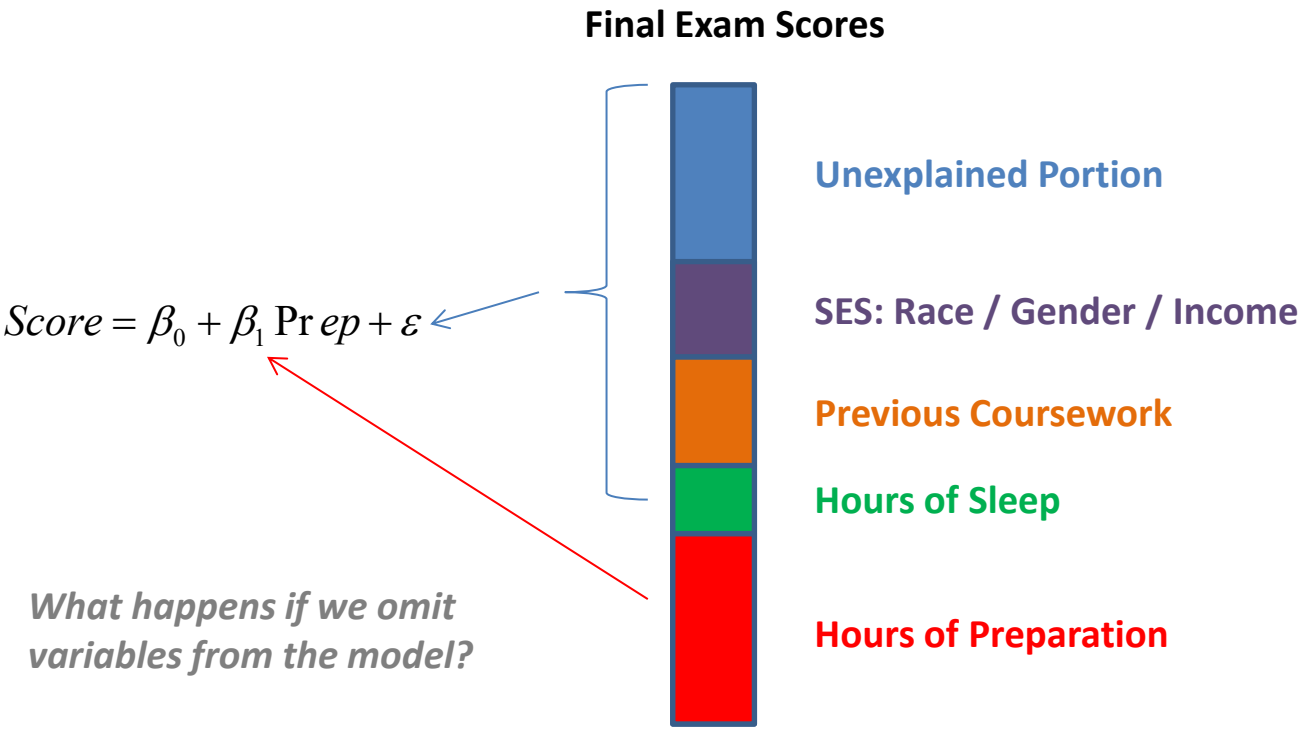
EFFECTS OF CONTROLS DETERMINED BY THEIR CORRELATIONS WITH THE OUTCOME AND POLICY VARIABLES



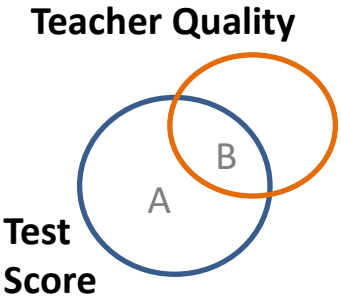


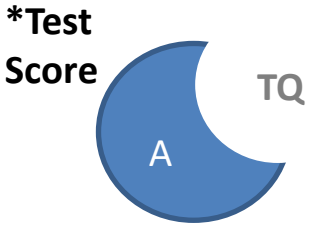


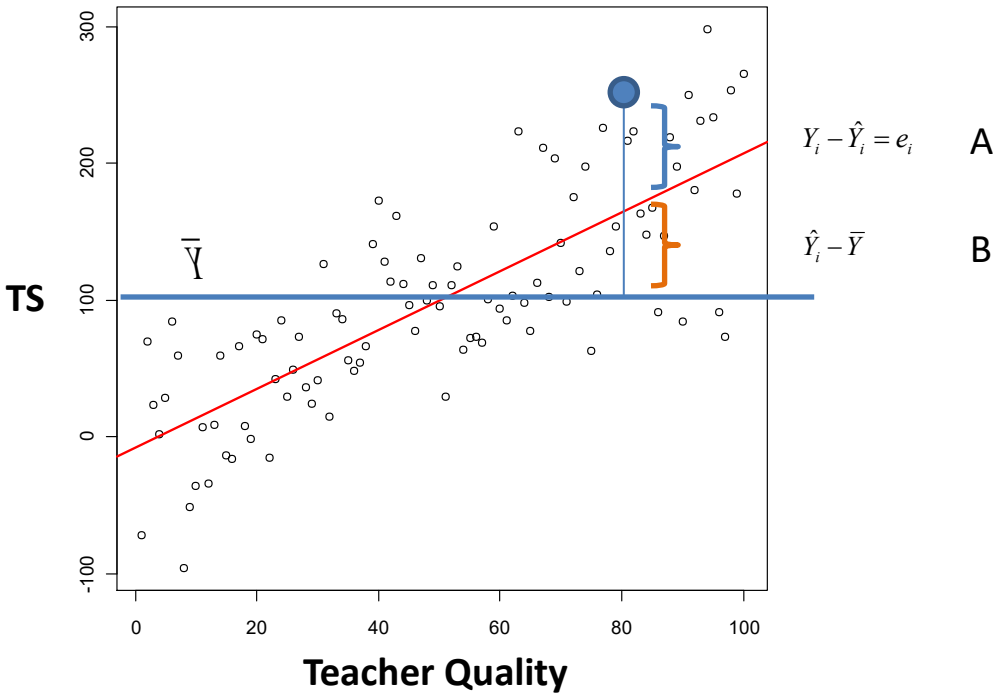
PARTITIONING THE VARIANCE OF Y



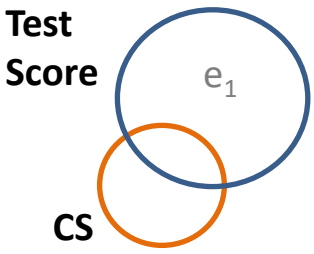
PARTITIONING THE VARIANCE OF Y

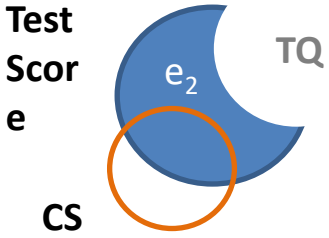


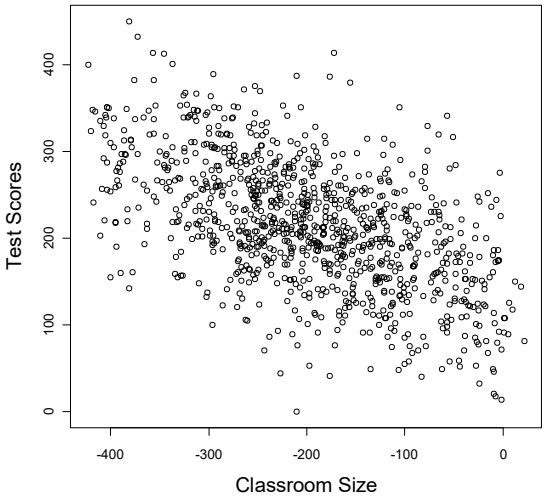


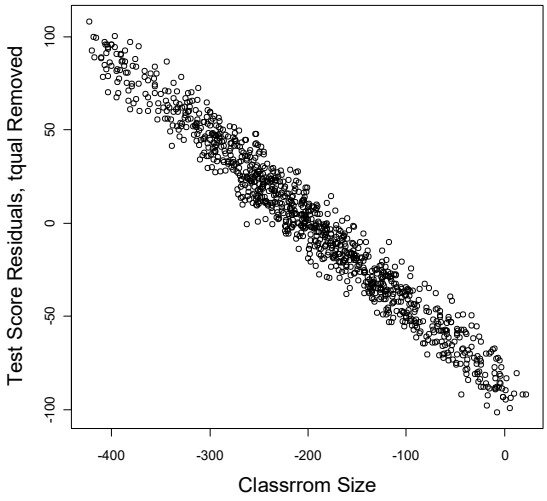


EFFECTS OF THE CONTROL VARIABLE

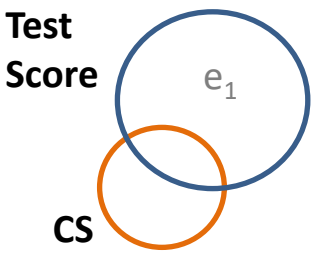




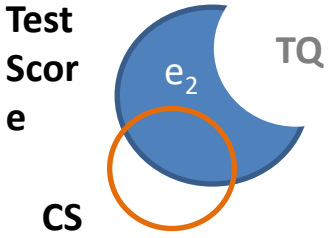


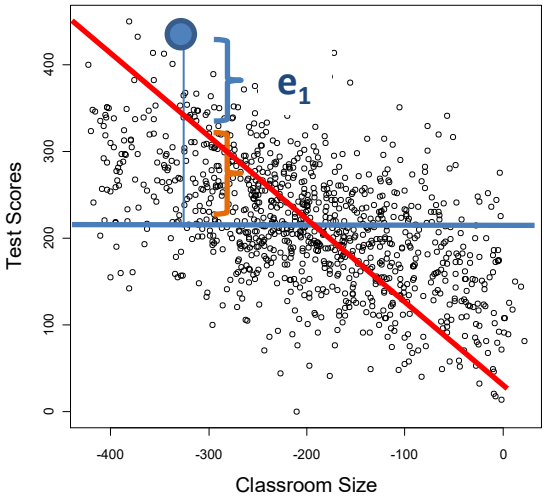


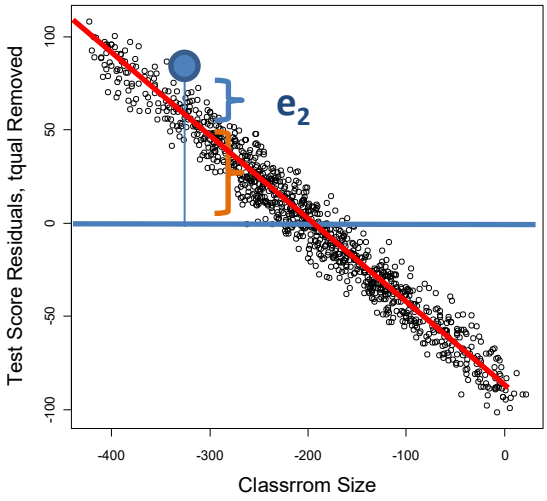
EFFECTS OF THE CONTROL VARIABLE



$e_1 > e_2$

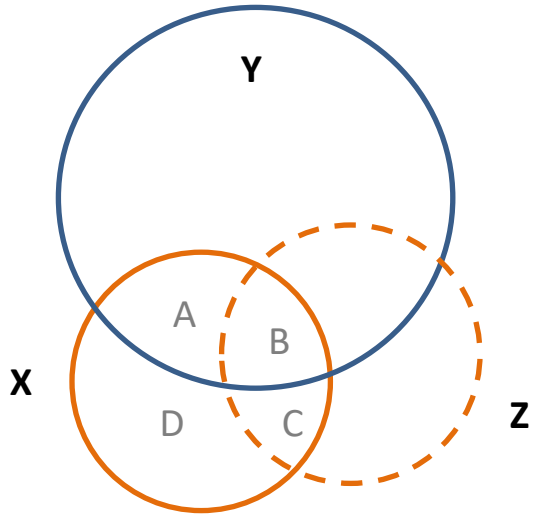






PARTITIONING Y WITH MULTIPLE VARIABLES

WHAT HAPPENS TO SLOPES AND STANDARD ERRORS WHEN WE ADD MORE VARIABLES?



MULTIPLE REGRESSION MECHANICS: SLOPE

$$Y = b_0 + b_1 X + e$$

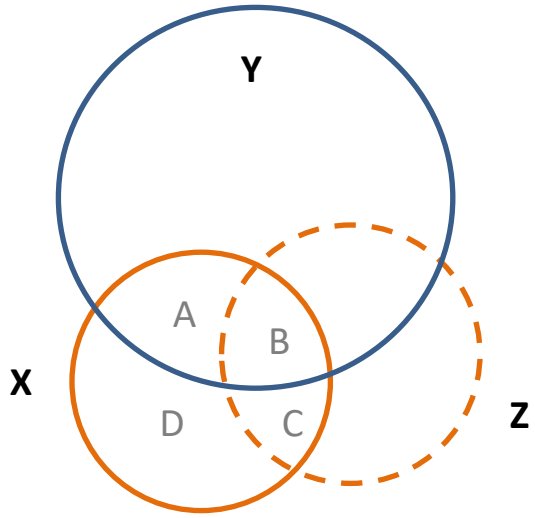
$$b_1 = \frac{\text{cov}(x, y)}{\text{var}(x)}$$

$$b_1 = \frac{A + B}{A + B + C + D}$$

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \varepsilon$$

$$\beta_1 = \frac{r_{yx} - r_{yx} r_{xz}}{1 - r_{xz}^2}$$

$$\beta_1 = \frac{A}{A + D}$$

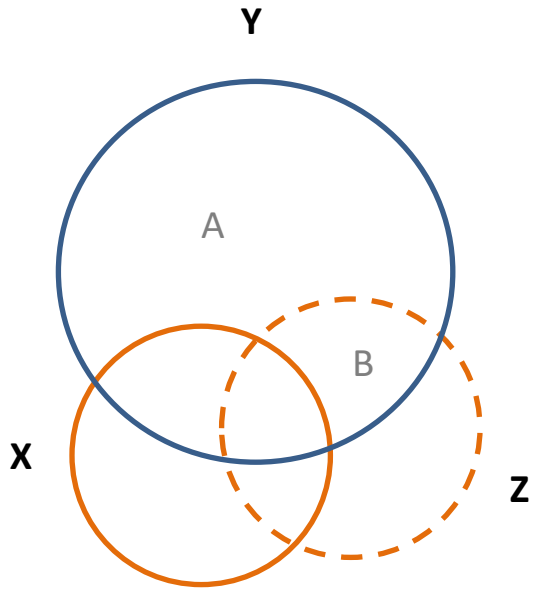


The formula for the slope is different when you add more variables

MULTIPLE REGRESSION MECHANICS: STANDARD ERRORS

$$Y = b_0 + b_1 X + e$$

$$e = A + B$$



$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \varepsilon$$

$$\varepsilon = A$$

The unexplained variance is different when you add more variables